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Enhanced Reserves Evaluation in Adsorbed Gas-Rich Unconventional Reservoirs Using a Modified Flowing Material Balance Approach: A Field Case Study-Based Practical Workflow

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Abstract

Accurate reserves evaluation is essential across all reservoir types, as it underpins effective development planning, informed investment decisions, and optimized field performance throughout the asset lifecycle. However, in unconventional reservoirs, ultra-low permeability, nonlinear flow behavior, and the presence of multiple gas storage mechanisms pose significant challenges to the reliability and applicability of conventional methods. This paper presents a novel flowing material balance (FMB) approach tailored for such reservoirs and compares it with various existing FMB-based techniques. A practical field case study is used to demonstrate how this semi-analytical workflow simplifies reserves estimation and maintains coherence with other established methods.

This methodology utilizes the Jensen-Smith Static Material Balance by incorporating dynamic production data and flowing pressure, eliminating the need for complex average reservoir pressure measurements. Isotherm-based sorption models are integrated with phase-specific productivity indices to develop a practical flowing material balance (FMB) framework suitable for field application. Applied to coalbed methane data with dual-porosity and high sorption capacity, the approach is validated using diagnostics such as P-Q plots, Normalized Gas Potential (NGP), Material Balance Time (MBT), the Normalized Rate-Cumulative Generalized FMB method, and other standard techniques, followed by analytical forecasting under various sensitivity scenarios.

The proposed flowing material balance (FMB) methodology demonstrated a significantly improved match between estimated and actual connected gas-in-place (GIP) in coalbed methane reservoirs with dominant sorbed gas storage. Multiple blind-tests and sensitivity analyses confirmed the method's robustness across varying reservoir conditions. By integrating sorption dynamics and flowing pressure into the FMB framework, the model effectively captured both early transient behaviour and late-time boundary-dominated flow, improving the reliability of reserves estimation. Comparative results showed connected GIP estimates within engineering margin of those obtained from King's method and the Generalized FMB, demonstrating consistency and validating methodological soundness. Field application confirmed its capability to characterize dynamic drainage areas, identify flow regimes, and assess well productivity trends with greater precision. Additionally, the methodology provided improved insight into completion efficiency

and depletion patterns, facilitating more informed decisions around well spacing, stimulation design, and asset development planning.

Keywords: Unconventional Reservoirs, Adsorbed Gas, CSG, Reserves Evaluation, Diagnostic Plots, Reservoir Surveillance, Field Development, Reservoir Management

Introduction

Global primary energy demand is expected to grow by approximately 20% between 2022 and 2050, driven by population growth, industrialization, and economic development in emerging markets. During this period, the global energy mix will undergo a significant transformation. Natural gas is projected to increase its share to nearly 26% of total energy consumption by 2050, with demand rising to approximately 5,360 billion cubic meters (bcm)—a 34% increase over 2022 levels ([Gas Exporting Countries Forum, 2024](#)). Notably, natural gas demand is not anticipated to peak within this timeframe. In parallel, renewable energy sources are expected to grow from a 3% share to over 17%, reflecting a broader shift toward low-carbon energy systems.

This evolving energy landscape underscores the strategic importance of unconventional hydrocarbon resources—including shale gas, tight gas, and coalbed methane (CBM)—in meeting long-term demand. These reservoirs are characterized by ultra-low matrix permeability, complex pore structures, and the presence of hydrocarbons in non-traditional forms such as adsorbed gas and high-viscosity fluids. As defined by ([Salazar Vanegas, 2007](#)), such accumulations are economically recoverable only through the application of advanced technologies, including horizontal drilling, multi-stage hydraulic fracturing, and specialized stimulation treatments.

Shales, while the most abundant sedimentary rock type in sedimentary basins, are only partially productive. A small subset possesses the necessary geomechanical and petrophysical properties for commercial gas production. Key evaluation parameters include total organic carbon (TOC), thermal maturity, porosity, permeability, free and adsorbed gas content, and rock mechanical properties.

CBM reservoirs, in contrast, store gas primarily through adsorption onto coal surfaces and release it via pressure-dependent desorption. Accurate reserves evaluation in CBM requires characterization of cleat systems, gas content, saturation, and Langmuir isotherm behavior. These parameters are essential for estimating gas-in-place (GIP) and forecasting production.

This paper addresses the technical complexities associated with reserves evaluation in unconventional reservoirs dominated by adsorbed gas. It begins by reviewing the fundamental storage mechanisms and reservoir characteristics, followed by a critique of conventional evaluation methods. The discussion then transitions to emerging techniques and integrated workflows designed to improve the accuracy and reliability of reserves estimation in these challenging systems.

Pore Architecture and Gas Storage Mechanisms in Coals and Shales

In unconventional gas reservoirs such as coalbed methane (CBM) and shale, gas is stored through a combination of adsorption, free gas accumulation, and solution gas in fluids or organic matter. CBM formations are characterized by a dual-porosity system, comprising a microporous organic matrix and natural fractures (cleats), where adsorption onto internal surfaces of organic matter dominates due to high surface areas often exceeding 100 m²/g ([C. R. Clarkson & Marc Bustin, 1996](#)).

In contrast, shale reservoirs exhibit more complex mineralogy and pore architecture, including intergranular and intragranular pores, microfractures, and nanoporosity associated with both organic matter and clay minerals ([C. R. Clarkson et al., 2013](#)). Pore sizes in shales typically fall within the micro- to mesopore range (<2-50 nm), as classified by the IUPAC system ([Thommes et al., 2015](#)), and their internal surface areas range from 2-20 m²/g, significantly lower than coals. Gas in shales is stored via adsorption on

kerogen and clays, free gas in matrix and fractures, and solution gas in fluids and bitumen (C. R. Clarkson, 2021).

The relative contribution of each mechanism depends on factors such as total organic carbon (TOC), thermal maturity, mineral composition, and reservoir pressure. Adsorbed gas is especially critical in low-permeability and low-porosity reservoirs, where it can constitute a substantial portion of the total gas-in-place (GIP). Accurate quantification of this component is essential for reliable reserves estimation and production forecasting, as it directly influences desorption-driven flow and recovery efficiency.

Gas adsorption in coal and shale reservoirs is governed by pore structure, surface area, and thermodynamic interactions between gas molecules and the solid matrix. The IUPAC classification system defines six types of adsorption isotherms, with Type I representing microporous solids—typical of coal and organic-rich shale—where monolayer adsorption dominates (Thommes et al., 2015).

In coalbed methane (CBM) systems, gas content is a non-linear function of pressure and is best described by Type I isotherms, indicating strong adsorption affinity even at low pressures. Shale reservoirs, while more mineralogically heterogeneous, also exhibit adsorption behavior influenced by pore size distribution and surface chemistry. Common models used to characterize adsorption include the Langmuir model for monolayer adsorption and the BET model for multilayer adsorption, both widely applied in shale gas studies (Andersen, 2021).

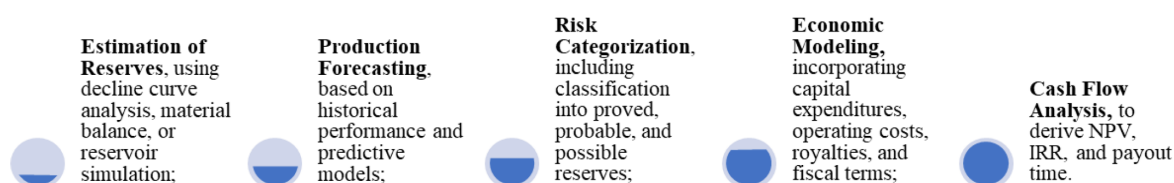
For multicomponent gas mixtures, the extended Langmuir model is frequently employed, incorporating mole fractions and component-specific parameters to estimate total adsorbed volume (C. R. Clarkson, 2003). Adsorption capacity is controlled by several reservoir-specific factors, including pressure, temperature, moisture content, mineralogy, organic matter type and maturity, and gas composition (Li et al., 2023). These variables influence both the quantity and behavior of adsorbed gas, which can constitute a significant portion of the total gas-in-place in such unconventional reservoirs. Accurate modeling of adsorption is essential for reserves estimation and production forecasting, particularly in systems where desorption-driven flow is a key recovery mechanism.

Reserves Evaluation Techniques

Reserves evaluation is a structured process used to quantify recoverable hydrocarbons and assess the economic potential of a reservoir. It integrates production forecasting, cost modeling, and financial analysis to estimate key metrics such as net present value (NPV), internal rate of return (IRR), and pay-out time. These outputs inform decisions related to field development, asset transactions, and investment planning.

Fundamentally, hydrocarbon volumes in a reservoir are classified into resources and reserves. Resources represent the total estimated quantity of oil or gas believed to be present, including volumes that may not yet be technically or commercially recoverable. Reserves, a subset of resources, refer specifically to volumes that have been demonstrated to be recoverable under existing economic and technological conditions. This distinction is critical for risk categorization and portfolio valuation.

Key components of a reserves evaluation include:



This paper primarily focuses on the first two essential steps in reserves evaluation: estimating the reserves and forecasting future production.

Performance-based Reserves Evaluation Methods

Performance-based methods—such as material balance, reservoir simulation, and production performance trend (PPT) analysis—are widely used to estimate technically recoverable resources (Harrell et al., 2004). PPT analysis, traditionally known as decline curve analysis (DCA), was formalized by Arps, 1945 and remains a reliable tool for production forecasting under stable flow conditions.

Modern PPT approaches also include rate transient analysis and production decline analysis, especially for unconventional resources. These methods extrapolate historical production data to estimate cumulative output at the economic limit, assuming stable lift mechanisms and well design.

PPT methods are sensitive to reservoir-specific factors, production practices, and facility constraints. Any significant changes can lead to inaccurate forecasts, making evaluator expertise and analog validation critical for dependable results.

	Pre-Drill	Post-Drill			
	Prospect	Explore	Appraise	Delineate	Produce
	Prospective Resources	Contingent Resources		Reserves	
Analogy	████████	██████	██████		
Volumetrics	████████	██████	██████		
Material Balance		██████	██████	██████	██████
Simulation		██████	██████	██████	██████
Decline Curve Analysis				██████	██████

In the early stages of reservoir evaluation, volumetric analysis and analogy methods are typically employed to estimate hydrocarbons initially in place (HIIP) and potential recovery. These approaches are particularly useful when limited dynamic data are available, relying instead on static geological and petrophysical inputs or analog field performance.

As more data become available—particularly from offset wells or existing geocellular models—reservoir simulation becomes increasingly applicable. Simulation is most effective when sufficient historical production and pressure data exist to enable a reliable history match, allowing for more accurate forecasting and scenario testing.

Material balance methods are best applied once adequate production history and at least one reliable measurement of static reservoir pressure are available. These methods provide a valuable cross-check on volumetric estimates and can be used to validate simulation results or test reservoir behavior under different development scenarios.

When historical pressure, production, and performance data are of sufficient quality, a combined approach—integrating reservoir simulation, material balance, and production performance trend (PPT) analysis (e.g., decline curve analysis)—can be used to estimate both recoverable volumes and hydrocarbons in place. This multi-method strategy enhances confidence in resource estimates and supports more robust development planning.

Scope and Methodology

This study focuses on an unconventional reservoir system with sufficient production history to support performance-based evaluation. However, the reservoir presents limited pressure control and considerable uncertainty in simulation parameters, including reservoir heterogeneity, pressure dependent static and dynamic properties, reservoir connectivity, etc. These factors limit the reliability of full-scale reservoir simulation at this stage.

Additionally, obtaining static reservoir pressure data is particularly challenging in low-permeability settings due to extended shut-in requirements and slow pressure equilibration. To address these constraints, this paper adopts a material balance approach, utilizing flowing pressure data under boundary-dominated flow conditions to estimate hydrocarbons in place and assess reservoir behavior.

While decline curve analysis (DCA) is a valuable tool in unconventional reservoir evaluation, it is not applied at this stage. Instead, the focus remains on dynamic material balance, which offers a more direct and physics-based interpretation of reservoir performance under the current data conditions.

The methodology developed here will be demonstrated using practical field data from a mature CBM field in India. However, it can also be applied solely to the calculation of adsorbed gas in some organic-rich shale gas reservoirs and similar unconventional settings.

A number of material balance equations (MBEs) have been developed specifically for coalbed methane (CBM) reservoirs, including those proposed by King, 1993, Jensen D.; Smith L. K., 1997, Seidle, 1999, Clarkson, 2001, Ahmed et al., 2006, Firanda, 2011, and Thararoop et al., 2015, among others. While all CBM MBEs account for gas desorption as a fundamental mechanism, they differ in their ability to incorporate free-gas storage and additional drive mechanisms such as water influx or compressibility effects.

C. R. Clarkson & Bustin, 2011 conducted comparative evaluations of various CBM MBEs and highlighted that while most methods perform adequately under idealized conditions, their accuracy and applicability can vary significantly with reservoir complexity and data quality. For saturated CBM reservoirs with low free-gas content and volumetric behavior, Clarkson's findings support the use of simpler MBEs—such as Jensen and Smith's—as they provide sufficiently accurate results without requiring extensive input data or complex modeling.

In this study, we adopt the **Jensen and Smith method** due to its **simplicity, minimal data requirements, and relevance to the reservoir conditions** under investigation. The reservoir is characterized as a saturated, two-phase CBM system with low free-gas storage and negligible water encroachment. Under these conditions, the Jensen and Smith method has been shown to yield gas-in-place and drainage-area estimates within acceptable engineering error margins, making it a practical and reliable choice for the current analysis.

Derivation of Jensen and Smith "Modified Material Balance (MMB)" Equation

This equation was derived using the simple mass balance statement provided below:

$$Gp = \text{OGIP} - \text{GIP} \quad (1)$$

Where Gp = cumulative gas production, Mscf, OGIP = original gas-in-place, Mscf, and GIP = current gas-in place, Mscf.

Substituting in GIP equation,

$$Gp = \left(\frac{43,560 \phi_f (1 - S_{wi})}{B_{gi}} + 1.3597 \left(\frac{V_L P_i}{p_i + p_L} \right) A_d h \rho_c \right) - \left(\frac{43,560 \phi_f (1 - S_{wi})}{B_g} + 1.3597 \left(\frac{V_L P}{p + p_L} \right) A_d h \rho_c \right) \quad (2)$$

If, $\frac{P_i}{p_i + p_L} \gg \frac{\phi_f (1 - S_{wi})}{B_{gi}}$ then the above equation can be re-arranged to derive the MMB:

$$\frac{P}{(P_L + P)} = -\frac{1}{1.3597V_L A_d h \rho_c} G_p + \frac{P_i}{(P_L + P_i)} \quad (3)$$

Where,

P = Average Reservoir Pressure within the drainage area, psia

P_L = Langmuir Pressure, psia

V_L = Langmuir Volume, scf/ton

P_i = Initial Reservoir Pressure, psia

ϕ_f = Fracture Porosity, fraction

S_{wi} = Initial Water Saturation in cleat/fracture, fraction

B_g = Gas Formation Volume Factor, rcf/Mscf

ρ_c = Bulk density of coal, g/cc

A_d = Drainage Area, acre

Derivation of the equation assumes a volumetric reservoir (no significant influx of water or change in reservoir volume), dominant adsorbed gas storage compared to free-gas storage, coals are saturated and sorption follows the Langmuir isotherm.

Uncertainty in OGIP estimation using this method may arise from inaccurate pressure data, the use of single-component Langmuir parameters in multi-component gas systems, non-volumetric reservoir behavior, and multilayer systems without crossflow—where high-permeability zones deplete faster than low-permeability ones. Additionally, significant free gas storage, if not properly accounted for, can further impact accuracy.

Generating a traditional material balance plot typically requires shutting in the well at multiple stages during its production life to measure average reservoir pressure. However, this approach is often impractical in low-permeability unconventional reservoirs due to operational constraints, and the shut-in durations are usually insufficient for accurate pressure readings. Instead, a Flowing Material Balance (FMB) method is employed, which leverages the principles of boundary-dominated flow and pseudo-steady state flow. In boundary-dominated flow portion of the constant pressure formulation, the pressure transient has reached all reservoir boundaries, and the static pressure begins to decline at the boundaries, though not uniformly due to variable flow rates. Pseudo-steady state flow portion of the equivalent constant rate formulation assumes a stabilized system where the pressure transient has reached all boundaries and the pressure declines uniformly throughout the reservoir.

Constant-rate and constant-pressure solutions serve as theoretical boundary conditions that real reservoir behavior tends to approximate but rarely matches exactly. During early production phases and structured production tests, reservoir performance often resembles the constant-rate condition, which is valuable for evaluating reservoir properties such as permeability and formation damage. As production progresses, reservoir behavior typically shifts toward the constant-pressure condition, which plays a key role in estimating the original volume of hydrocarbons in place and the amount of reserves that can be recovered.

Maintaining a constant production rate is operationally challenging; therefore, (Mattar et al., 2006) proposed a dynamic material balance equation for variable-rate cases.

$$\frac{m(P_i) - m(P_{wf})}{q_g} = -\frac{t_{ca}}{G_i C_{ti}} + \frac{(\mu B)_i}{(2\pi K h)} \left[\frac{1}{2} \ln \left(\frac{4A}{C_A e^{\gamma} r_w^2} \right) \right] \quad (4)$$

$$\frac{\Delta m(P)}{q_g} = m_a t_{ca} + b_{pss} \quad (5)$$

$$\Delta m(P) = m(P_i) - m(P_{wf}) \quad (6)$$

$$m_a = \frac{1}{G_i C_{ti}} \quad (7)$$

$$b_{pss} = \frac{(\mu B)_i}{(2\pi K h)} \left[\frac{1}{2} \ln \left(\frac{4A}{C_A e^{\gamma} r_w^2} \right) \right] \quad (8)$$

$$t_{ca} = \frac{(\mu_g c_g)_i}{q_g} \int_0^t \frac{q_g}{(\mu_g c_g)} dt \quad (9)$$

$m(P') =$ Normalized pseudo-pressure, $\frac{psi^2}{cp}$

$\gamma =$ Euler's constant = 0.57721

$K =$ Permeability, mD

$h =$ Net Reservoir Thickness, ft

$A =$ Drainage Area, acre

$C_A =$ Reservoir-shape constant

$r_w =$ Wellbore radius, ft

During pseudosteady-state or boundary-dominated flow in a gas well, the total pressure drop from the initial reservoir pressure to the bottomhole flowing pressure comprises two components: the pressure drop due to depletion, $\frac{t_{ca}}{G_i C_{ti}}$, and the pressure loss due to inflow, that is $b_{pss} = \frac{(\mu B)_i}{(2\pi K h)} \left[\frac{1}{2} \ln \left(\frac{4A}{C_A e^{\gamma} r_w^2} \right) \right]$

Thus, during boundary dominated flow:

$$m(\bar{P}) = m(P_{wf}) + q_g b_{pss} \quad (10)$$

Given that the flow rate is known, the only remaining requirement is to determine the value of b_{pss} using an independent method to obtain an estimate for the average reservoir pressure.

Productivity Index (J)

The productivity index of a well is a measure of the production rate achievable under a given drawdown pressure, which is the difference between the average reservoir pressure and the flowing bottomhole pressure. Defined as the flow rate per unit pressure drop, the productivity index gives an indication of the production potential of a well.

$$J = \frac{1}{b_{pss}} \quad (11)$$

Using [equation 10](#), we obtain:

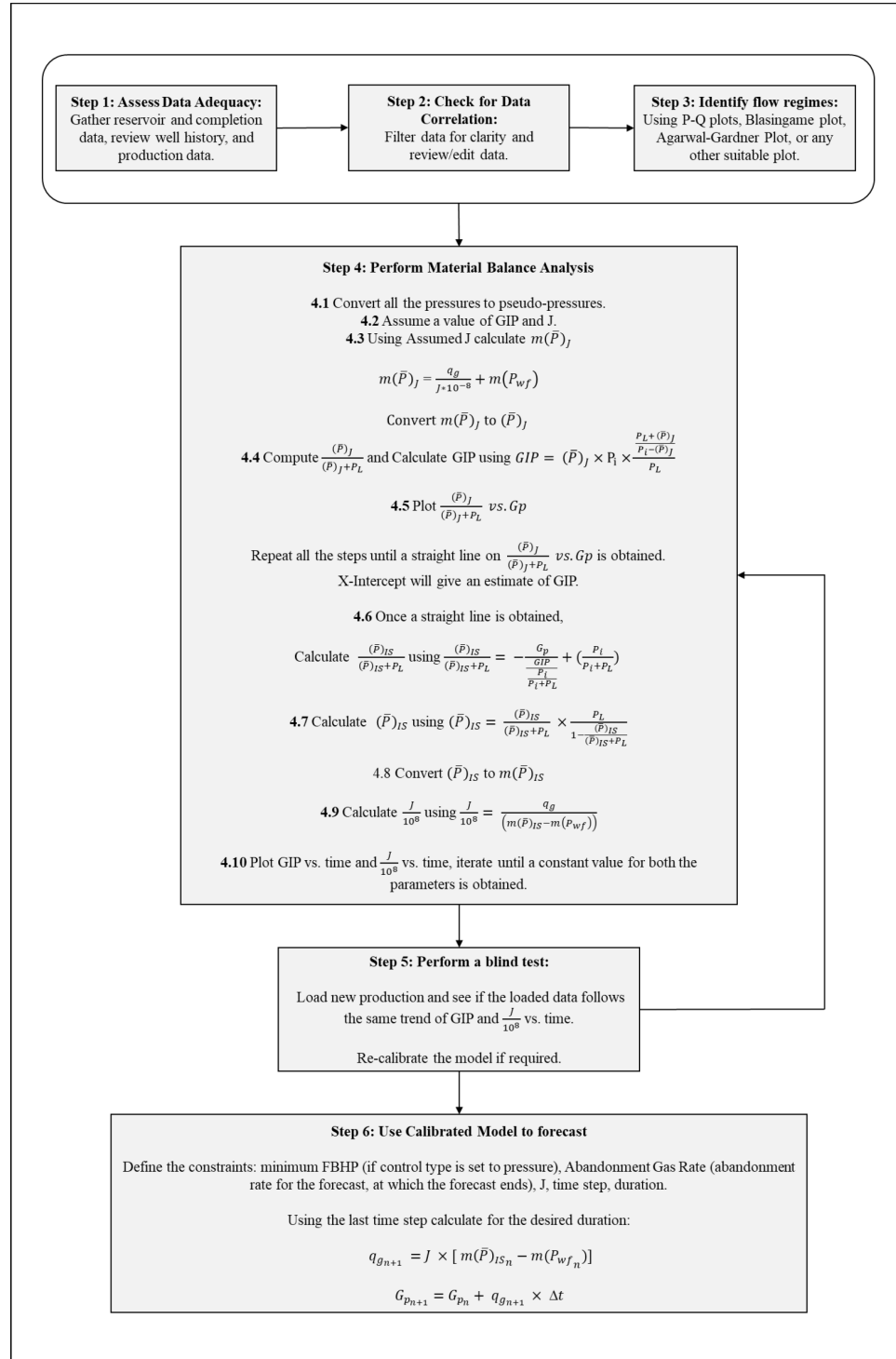
$$J = \frac{q_g}{m(\bar{P}) - m(P_{wf})}, \frac{MMSCFD}{\frac{psi^2}{cp}} \quad (12)$$

Using a plot of the Productivity Index versus Cumulative Gas Production: If the correct OGIP is calculated from the material balance analysis, the productivity index for the boundary-dominated portion of the data should plot as a flat line.

A downward sloping line can be an indication that the calculated volume is too large, while an upward sloping line can indicate that the calculated volume is too small.

Productivity indexes that fall off a flat line can also be an indication of operational issues with the well.

Coupling Modified Material Balance Equation and Dynamic Material Balance Equation



Where,

$q_{g,n+1}$ = Calculated Gas Rate at time step $n + 1$, MMSCFD

J = Gas Productivity Index, $\frac{\text{MMSCFD}}{\frac{\text{psi}^2}{\text{cp}}}$

$m(\overline{P})_{IS_n}$ = Pseudo — pressure normalized average reservoir pressure at time step n, $\frac{\text{psi}^2}{\text{cp}}$

$m(P_{wf_n})$ = Pseudo — pressure normalized flowing bottomhole pressure at time step n, $\frac{\text{psi}^2}{\text{cp}}$

$G_{p_{n+1}}$ = Cumulative Gas Production at time step n + 1, MMSCF

G_{p_n} = Cumulative Gas Production at time step n, MMSCF

Δt = time step, days/months/years

Similarly obtain $(\overline{P})_{IS}$ using step 4.7 as described previously to obtain the estimate of average reservoir pressure at the end of forecast. $G_{p_{n+1}}$ at the end of forecast duration gives the Estimated Ultimate Recovery (EUR).

Validation

The subsequent phase of the study involved validating the analytical material balance technique by comparing its performance against simulated production and depletion data. A Numerical Single Type-Well model, calibrated using field performance data and a predefined gas-in-place (GIP) value as input, was utilized. The objective was to evaluate the accuracy of the connected GIP and other parameters estimated through the analytical flowing material balance method by benchmarking them against the known outputs from the simulation model.

Table 1 summarizes the key inputs into the numerical model.

Table 1—Numerical Single Type-Well Model: Key Inputs

Top Depth	1935	ft
Initial Reservoir Pressure	148	psia
Net Thickness	16.4	ft
Cleat Permeability	14	mD
Cleat Porosity	0.015	fraction
Coal Gas Saturation	1	fraction
Gas-In Place	1040	MMSCF
Drainage Area	260	Acre

Using the developed workflow, simulated data from the initial 11 months were utilized to construct the material balance plot (Fig. 1) and estimate key reservoir parameters, including connected GIP (Fig. 2), drainage area, and average reservoir pressure (Fig. 4) at the end of the forecast period.

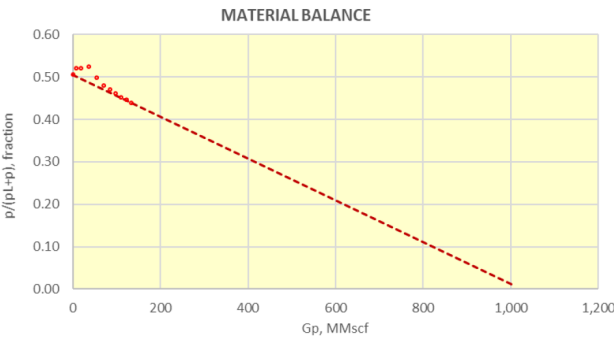


Figure 1—Material Balance Plot of Simulated Data

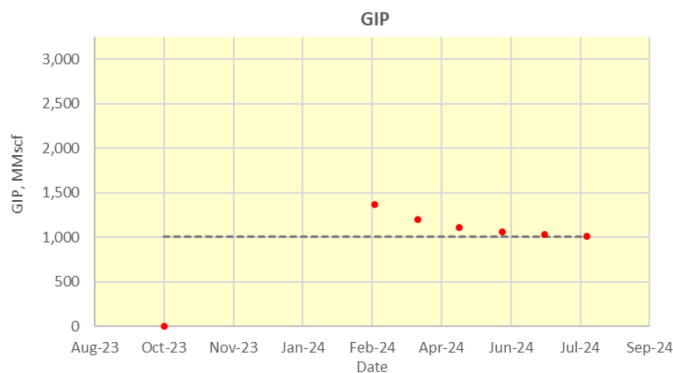


Figure 2—Calculated GIP (MMSCF) using Modified FMB Approach

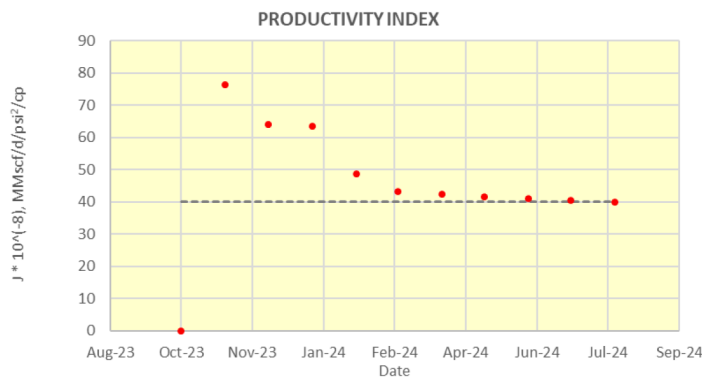


Figure 3—Calculated PI using Modified FMB Approach

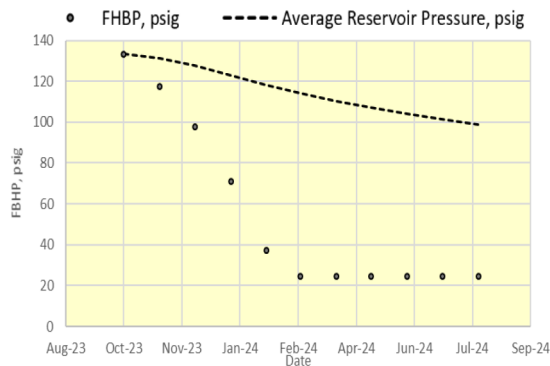


Figure 4—Calculated Average Reservoir Pressure (psig) using Modified FMB Approach

From the material balance analysis, a connected GIP of 1,010 MMSCF, a Gas Productivity Index of $40 \times 10^{-8} \frac{\text{MMSCFD}}{\frac{\text{psi}^2}{\text{cp}}}$, and a drainage cp area of 251 acres were derived, which are within the engineering margin of error when compared to the numerical simulation model. Using the forecasting methodology outlined earlier, a 30-year production forecast was conducted, yielding estimates for GIP (Fig. 5) and average reservoir pressure (Fig. 6) at the end of the forecast period.

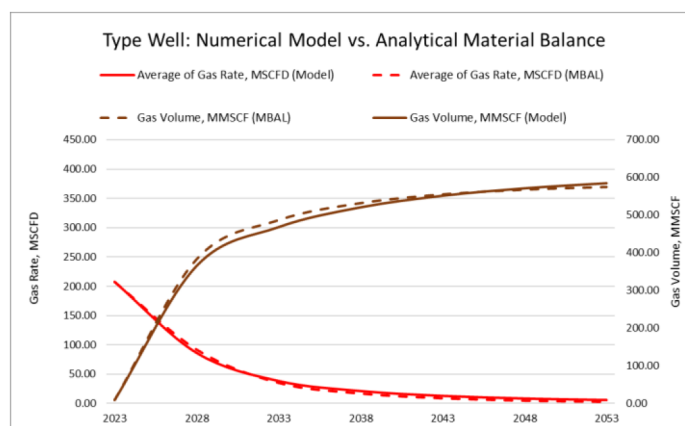


Figure 5—Gas Rate (MSCFD) and Gas Volume (MMSCF) Comparison: Numerical Model vs. Analytical Material Balance

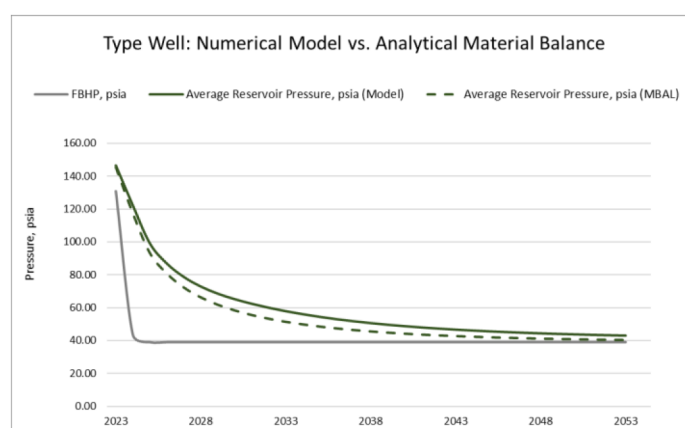


Figure 6—Average Reservoir Pressure (psia) Comparison: Numerical Model vs. Analytical Material Balance

Table 2 presents a comparative summary of the key parameters derived from the numerical simulation model and the analytical material balance approach.

Table 2—Comparative Summary - Numerical Simulation Model and the Analytical Material Balance Approach

Parameter	Numerical Model (Single Type-Well)	Analytical Model (Material Balance)	Absolute Error / % Error
GIP, MMSCF	1040	1010	-30 / -2.88%
Drainage Area, Acre	260	251	-09 / -3.48%
30 Year EUR, MMSCF	585	575	-10 / -1.77%
Average Reservoir Pressure at the end of forecast, psia	43	40	-03 / -6.49%

The fundamental difference between numerical simulation and analytical material balance models lies in how each represents reservoir behavior and computes key parameters. Numerical simulation discretizes the reservoir into a grid and solves flow equations over time, capturing spatial and temporal variations in pressure, saturation, and fluid properties. This allows it to model heterogeneity, complex boundary conditions, and transient flow regimes with high fidelity. In contrast, the analytical flowing material balance method simplifies the reservoir to a single homogeneous tank and estimates parameters like gas-in-place (GIP) and average pressure using bulk production data and the real gas law, incorporating compressibility. While simulation-derived average pressure is a spatial arithmetic mean, material balance pressure is a

volumetric average that reflects the pressure required to contain the same total mass in a single tank. These two approaches yield similar results only in ideal, homogeneous systems; in practice, due to pressure and composition gradients, they will differ—often meaningfully.

Field Case Study

Case 1: Horizontal well completed in a single coal seam

Well A is a horizontal well completed within one of the coal seams of the Sohagpur Field, India. Fig. 7 illustrates the well performance over time, including gas production rate, water production rate, and flowing bottom-hole pressure (FBHP). The methodology developed in this study was applied to estimate both the Connected Gas-in-Place (GIP) and the Estimated Ultimate Recovery (EUR).

The initial step involves diagnosing the flow regime. The P-Q diagnostic plot, as proposed by (Kabir & Izgec, 2006), provides a simplified approach to identifying boundary-dominated flow (Patel et al., 2025). A positive slope in the plot indicates boundary-dominated behavior, with the slope being inversely proportional to the connected reservoir volume (Fig. 8). Under conditions of constant flowing pressure, a zero slope is observed, signifying stabilized flow. Further validation of the flow regime is provided by the Water P-Q diagnostic plot (Fig. 9) and the observation of a constant water-gas ratio (Fig. 10), both of which support the boundary-dominated flow interpretation.

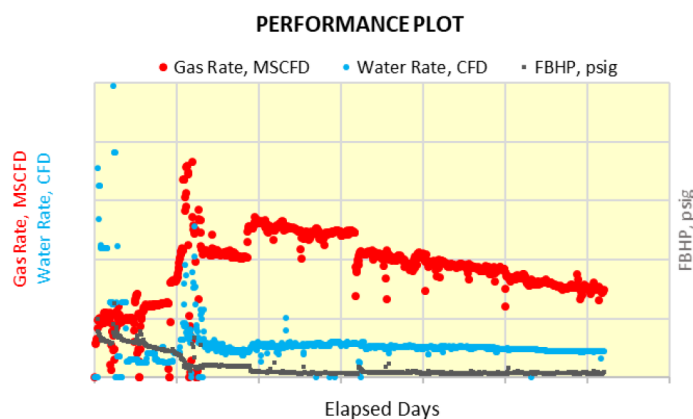


Figure 7—Performance Plot (Historical Production Data)

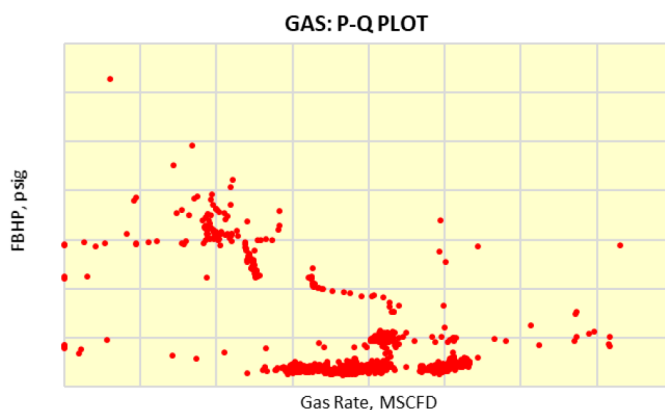


Figure 8—Gas P-Q Diagnostic Plot

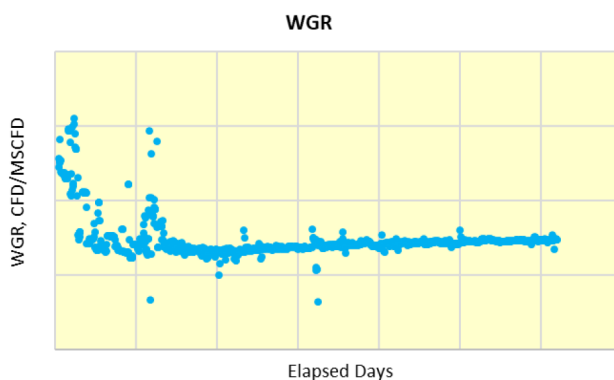


Figure 9—Water Gas Ratio vs. Elapsed Days

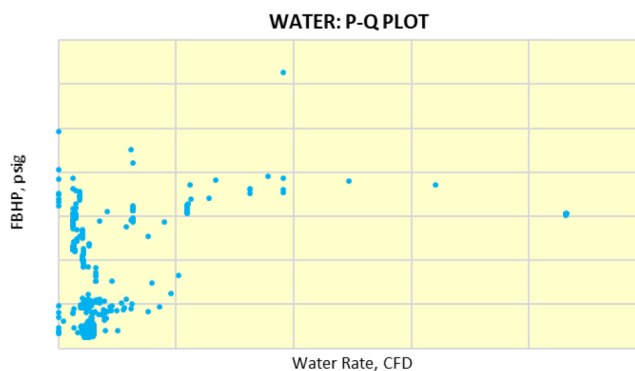


Figure 10—Water P-Q Diagnostic Plot

Following the proposed methodology, the Connected Gas-in-Place (GIP) (Fig. 11) and gas productivity index (PI) (Fig. 12) were determined iteratively until a linear trend was observed when plotted against elapsed production days. The average reservoir pressure, estimated using the PI, was employed to construct the material balance plot (Fig. 13). This plot was further refined by incorporating cumulative production data to correct the average reservoir pressure (Fig. 14).

Once a consistent linear trend was achieved on the material balance plot, the X-intercept of the plot was used to calculate the final GIP estimate, representing the total connected gas volume in the reservoir.

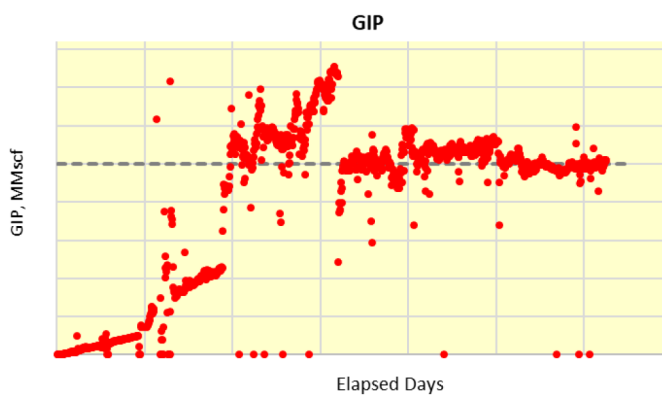


Figure 11—Calculated GIP (MMSCF) using Modified FMB Approach

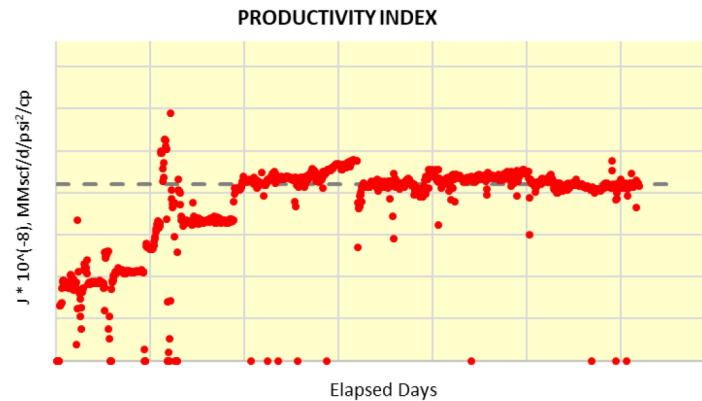


Figure 12—Calculated Gas PI using Modified FMB Approach

In CBM reservoirs, gas PI typically evolves over time (Fig. 12) due to dynamic interactions between increased free gas saturation and relative permeability near the wellbore increasing the gas PI, counteracting effects include cleat compaction from pressure depletion and matrix shrinkage from desorption, which respectively reduce and increase cleat aperture and thus the gas PI. These geomechanical and flow mechanisms vary temporally and requires rigorous quantification for detailed characterization.

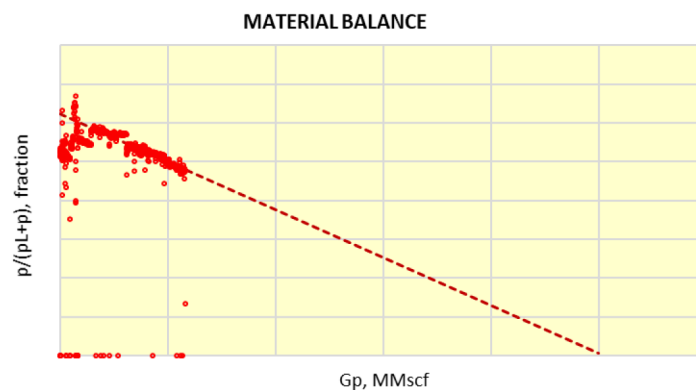


Figure 13—Calculated Material Balance Plot using obtained GIP and PI

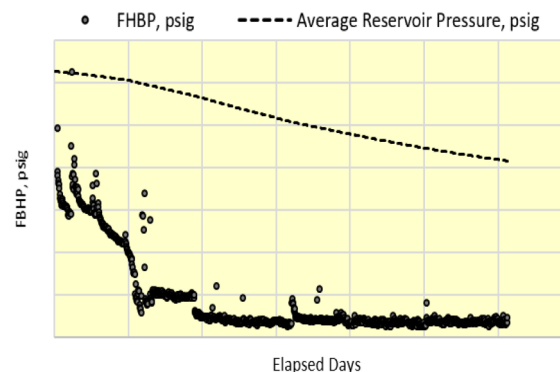


Figure 14—Calculated Average Reservoir Pressure using obtained GIP and PI

A blind validation test was conducted to verify the robustness of the estimated Connected Gas-in-Place (GIP) and productivity index (J). The initial 11 months of production data were utilized to derive unique values for GIP and J. These parameters were then used to forecast future performance, and the actual production data were compared against the forecasted values.

Fig. 15 presents a consistent match between the observed gas production rates and those predicted by the blind test, demonstrating the reliability of the derived parameters over time.

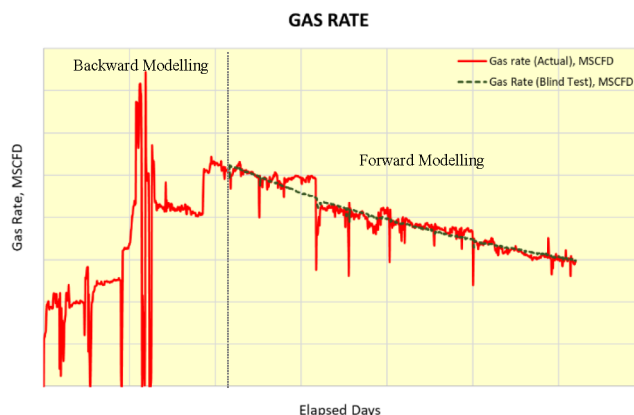


Figure 15—Blind Test to validate the obtained PI and GIP

The calibrated values of productivity index (J) and Gas-in-Place (GIP) were employed to generate the production forecast, from which the 30-year Estimated Ultimate Recovery (EUR) (Fig. 16) and the average reservoir pressure (Fig. 17) at the end of the forecast period was determined.

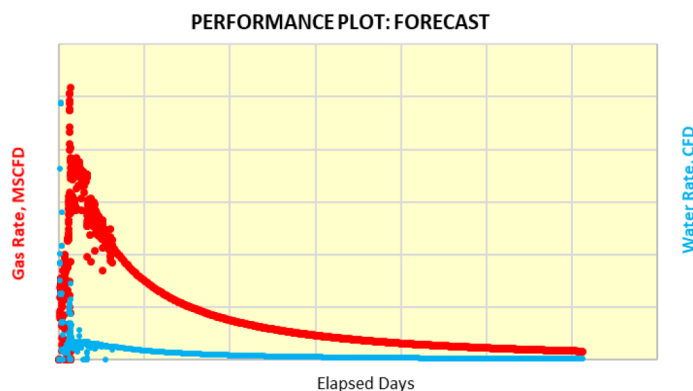


Figure 16—30 Years Production Forecast

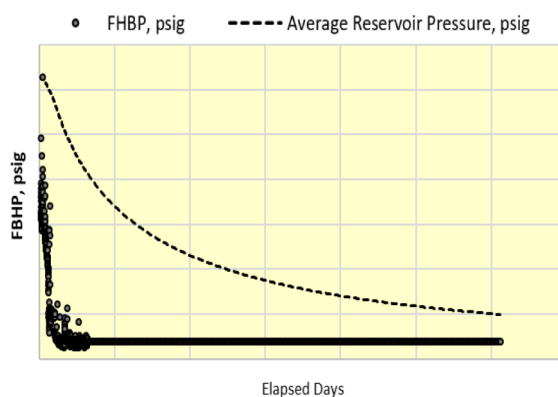


Figure 17—30 Years Pressure Forecast

Case 2: Multilateral well completed in a single coal seam

Well B, a planar multilateral well in the Sohagpur Field, India, was evaluated using the proposed methodology. Fig. 18 presents its production performance—gas rate, water rate, and FBHP—over time.

The analysis yielded estimates of Connected Gas-in-Place (GIP) and 30-year Estimated Ultimate Recovery (EUR).

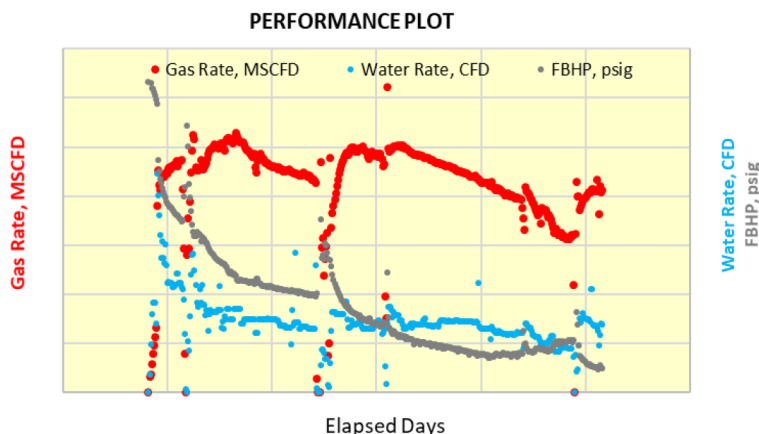


Figure 18—Performance Plot (Historical Production Data)

Given the reservoir heterogeneity and interconnectivity uncertainty, a rigorous diagnostic approach was employed to identify the flow regime. As shown in Fig. 19, normalized gas potential plotted against material balance time exhibited a unit slope during late-time, indicating the onset of boundary-dominated flow.

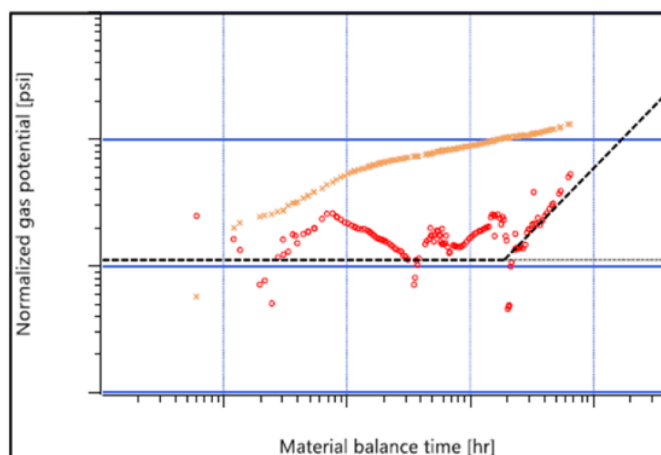


Figure 19—Normalized Gas Potential (psi) vs. Material Balance Time (hr) Plot

To assess the robustness of the proposed methodology, key parameters derived from the modified material balance analysis were benchmarked against established techniques. These included the flowing p/Z^{**} method, which accounts for multiple energy sources such as formation compressibility, residual fluid expansion, aquifer support, reservoir connectivity, and adsorption effects in coal/shale. Additionally, the Agarwal-Gardner (A-G) normalized rate (Agarwal et al., 1999) method was applied, involving inverse pseudo-steady state plotting of normalized rate versus normalized cumulative production. The x-intercept from the best-fit line was used to estimate Connected GIP, enabling cross-validation of results.

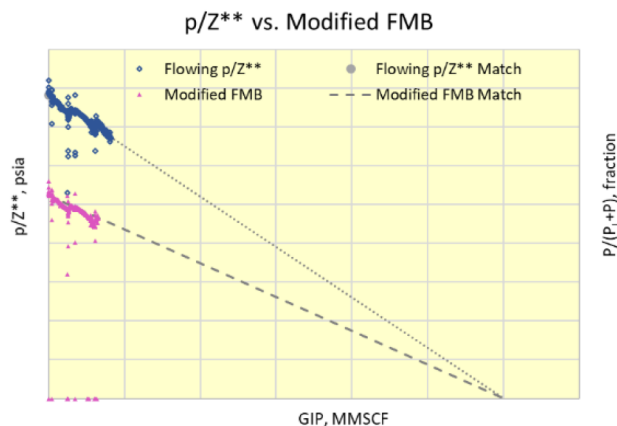


Figure 20—Flowing p/Z^{**} vs. Modified FMB: GIP (MMSCF) Comparison

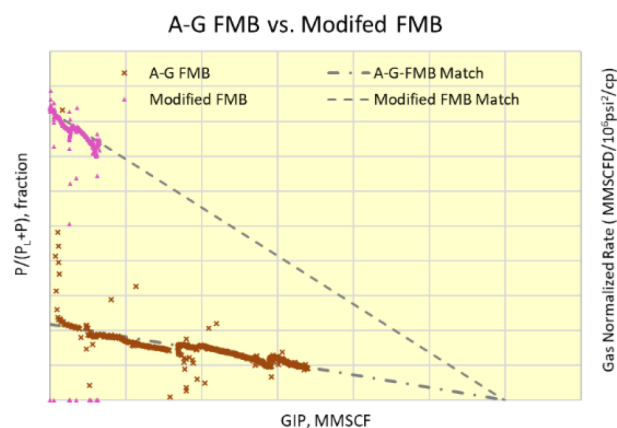


Figure 21—A-G FMB vs. Modified FMB: GIP (MMSCF) Comparison

Fig. 20 and Fig. 21 present a comparative analysis of Connected GIP estimates obtained from the flowing p/Z^{**} method, Agarwal- Gardner (A-G) normalized rate, and the modified material balance approach. The results demonstrate strong agreement across all methods, confirming the reliability of the proposed methodology. Additionally, the analytical model-based productivity index and reservoir pressure obtained from a commercial RTA software were compared with results from the modified flowing material balance approach. Fig. 22 and Fig. 23 highlight the contrast between these methods and demonstrate strong consistency in the estimated parameters.

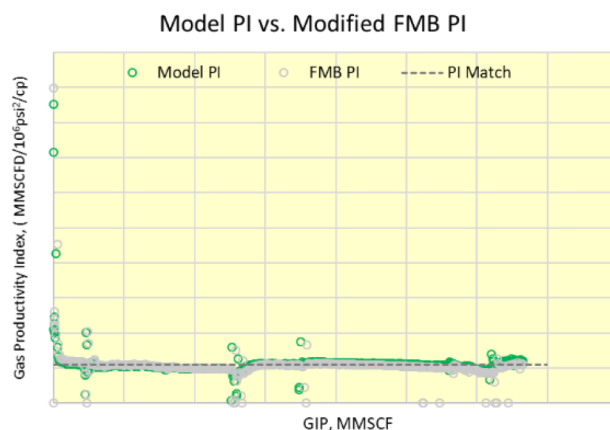


Figure 22—Analytical Model vs. Modified FMB: PI Comparison

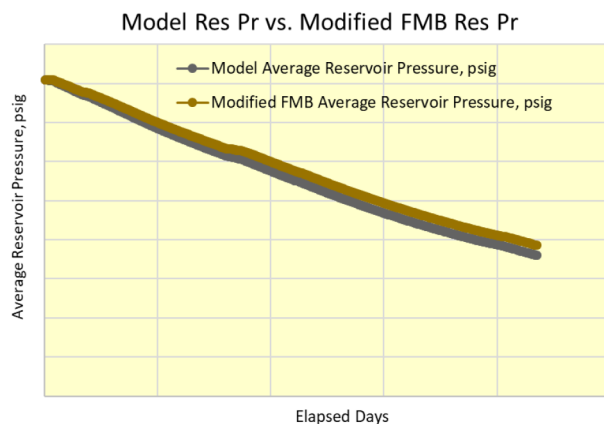


Figure 23—Analytical Model vs. Modified FMB: Average Reservoir Pressure (psig) Comparison

Finally, the calibrated values of productivity index (J) and Connected Gas-in-Place (GIP) from the modified material balance model were used to generate the production forecast. From this, the 30-year Estimated Ultimate Recovery (EUR) (Fig. 24) and the average reservoir pressure at the end of the forecast period (Fig. 25) were determined.

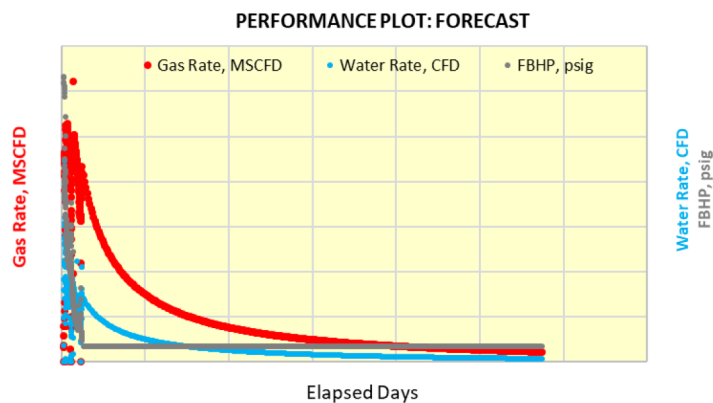


Figure 24—30 Years Production Forecast

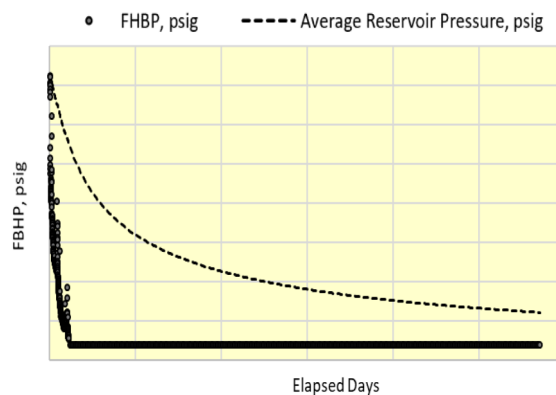


Figure 25—30 Years Pressure Forecast

Conclusion

Modified Flowing Material Balance (FMB) Approach

A novel approach was successfully implemented in adsorbed gas-rich unconventional reservoirs, particularly coalbed methane (CBM), enhancing the traditional static material balance method by integrating the flowing material balance approach.

It eliminates the need for static reservoir pressure measurements, making it more practical for low-permeability reservoirs.

Enhanced Accuracy and Reliability

The proposed method demonstrated strong accuracy in estimating Connected Gas-In-Place (GIP), yielding results within the engineering tolerance of established techniques such as Flowing p/Z^{**} method and the Generalized FMB (A-G FMB) approach.

Blind validation tests and sensitivity analyses across diverse scenarios confirmed the method's robustness and adaptability.

Integration of Sorption Dynamics

Incorporating Langmuir isotherm-based adsorption models improved the understanding of gas storage and flow behavior.

This integration allowed better modeling of both early transient and late boundary-dominated flow regimes.

Enhanced Reservoir Characterization

The workflow enabled precise estimates of dynamic drainage areas and well productivity trends.

It provided insights into depletion patterns, aiding in well spacing and development strategy.

Validation Against Simulation

Analytical results closely matched numerical simulation outputs for GIP, drainage area, and reservoir pressure.

Demonstrated that the analytical model is a viable alternative within engineering margin to full-scale simulation in data-constrained environments.

Field Applicability

Successfully applied to CBM field data in India, including horizontal and multilateral wells.

Forecasting results supported long-term production planning and reserve estimation.

Contextual Boundaries

While the modified FMB approach demonstrates strong alignment with simulation results and offers practical advantages in data-constrained environments, its effectiveness is most pronounced under conditions where reservoir behavior approximates volumetric assumptions, adsorbed gas dominates over free gas, and sorption follows Langmuir-type isotherms.

The accuracy of reserves estimation may be influenced by the quality of pressure data, the representativeness of single-component sorption parameters in multi-component systems, and the degree of reservoir heterogeneity.

In systems with layered architecture or uneven permeability distribution, flow dynamics may deviate from idealized assumptions, potentially affecting the precision of gas-in-place estimates.

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